Syracuse University
School of Information Studies

Week 1
Introduction to Data Science

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Syllabus Copy

- You must own a functioning laptop with R and RStudio installed or create an account on posit.cloud.
- You must bring your laptop to every class meeting, both lecture and lab.
- The course will consist of one larger class plus one smaller lab section each week.
- There will also be an **exam.**
- There will also be a **final project (group)** which you will complete during the semester.

Grading

- **Homework (24%):** The homework assignments are each worth **2 points**.
- **Weekly (quiz-based) Challenges (12%):** You need to complete these quizzes during each class meeting.
- **Lab Assignments (12%):** You will receive one point for your submission.
- **Exam (24%):** The exam is designed to take 80 minutes to complete. It will be open-book and open-notes, but you may not discuss the exam with other students.
- **Final Project (25%):** This will be completed during the last half of the course. More information will be made available during the course.
- **Participation (3%):** Includes professionalism in class and attending class.

Data Analytics v/s Data Science

Data Analytics

- Looking at **existing data** to understand **what happened and why**.
- In simple words:
  “Tell me what the data says about the past and present.”

What data analysts do:

- Clean data
- Make charts and dashboards
- Find trends and patterns

Tools commonly used:
Excel, SQL, Tableau, Power BI, basic Python/R

Data Science

- Using data to **predict the future** and **build intelligent systems**.
- **What data scientists do:**
  - Everything data analysts do **plus** build predictive models
  - Use machine learning and statistics
- **Tools commonly used:**
  Python, R, machine learning, deep learning, statistics, big data tools
- **Focus:**
  - Prediction and automation
  - Creating models and algorithms
- **One-line difference**
  - **Data Analytics:** *Understand the past*
  - **Data Science:** *Predict the future*

Structured & Unstructured Data

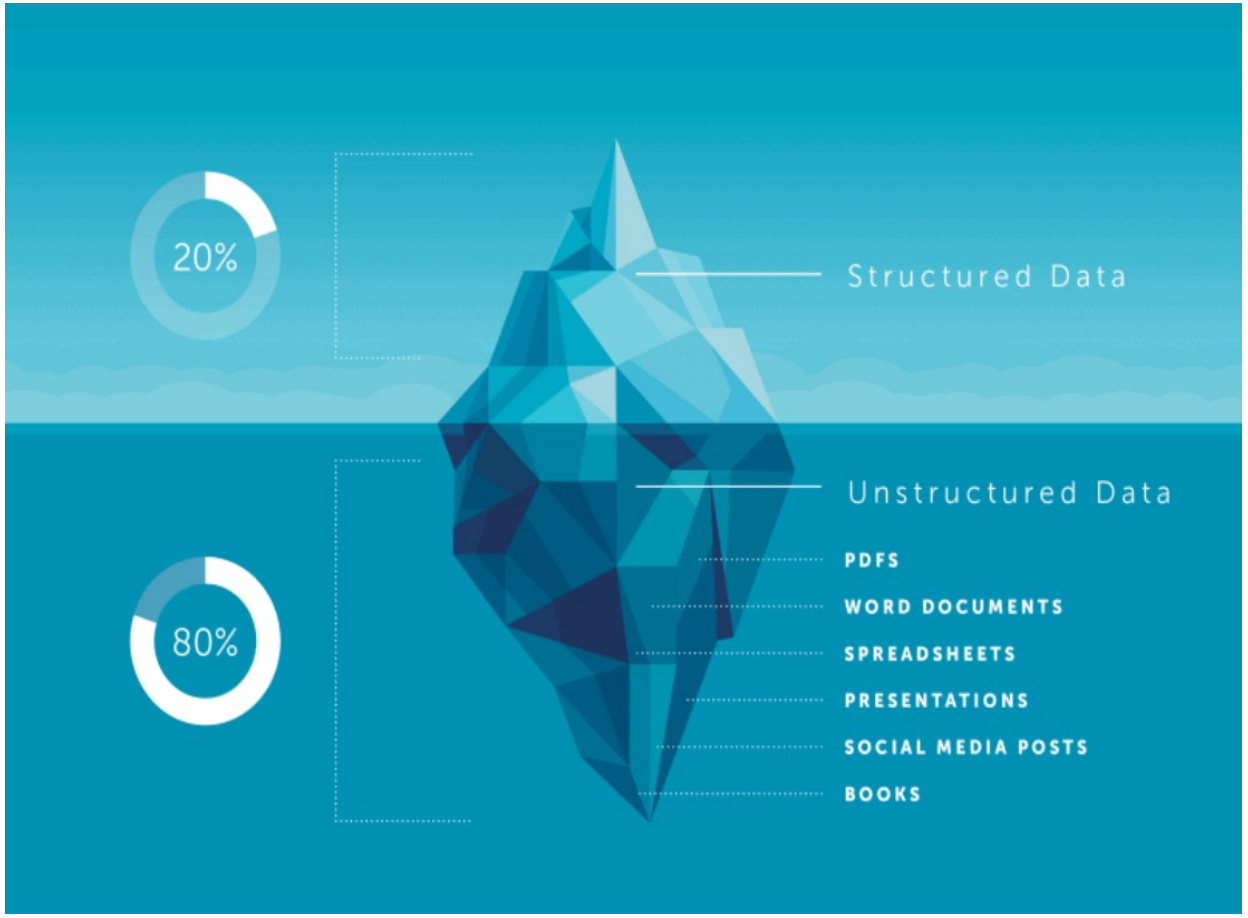
- **Structured Data:**
  - Organized and easy to search.
  - It's usually stored in tables with rows and columns, like a spreadsheet.
  - Computers can easily process it.

- **Unstructured Data:**
  - Not organized in a fixed way and doesn't fit neatly into tables.
  - Often more difficult for computers to search or analyze.
  - **Examples**: Social media posts, customer reviews, etc.
  - It might have useful information, but it's hidden inside messy data.
  - You often need special tools or techniques to understand it.

Question

Do data scientists mainly deal with structured data or unstructured data?

Answer


20%
Structured Data
80%
Unstructured Data
PDFS
WORD DOCUMENTS
SPREADSHEETS
PRESENTATIONS
SOCIAL MEDIA POSTS
BOOKS

Real World Applications

- Business Analytics
- Climate Analytics
- Health Analytics

Data Science Pipeline

- A simple, structured way to understand how data scientists solve real-world problems.
- The **O.S.E.M.N. Framework**: Each letter stands for a stage in the data science workflow.
- **O – Obtaining Data**:
  - **Identifying data:** Where can I get the data?
  - This is the first step, and it's all about collecting information from the right sources.
- **S – Scrubbing the Data (Cleaning):**
  - Is this data clean, consistent, and usable?
  - Real-world data is messy. It may have: Missing values, Typos, Duplicates, Wrong formats (like dates in weird formats)
  - Here, the goal is to clean up the data so it's reliable and ready for analysis.
  - This step often takes the most time (sometimes 60–70%).

- **E – Exploring and Visualizing:**
  - This is where you understand your data.
  - You use: Graphs (bar charts, scatter plots, histograms), Summary statistics (mean, median, etc.).

- **M – Modeling the Data:**
  - Now you use machine learning or statistical models to Predict a future event.
  - It only works well if the earlier steps were done right.

- **N – Interpreting the Results:**
  - A great model is useless unless someone can understand and use the results.
  - A data scientist has to explain insights clearly to people who are not technical, like business managers, doctors, or teachers.

- OSEMN iterative! You'll often go back and forth between steps.

- **Example:** After modeling, you might realize you need more data (go back to "O"), or you see strange results, so you revisit data cleaning ("S").

Data Science: Getting Started with R

What is R?

- R is a computer program that helps people work with data.

Data Science: Getting Started with R

Why is R special?

- Free
- Open-source: It's built and maintained by a big community of volunteers.
- Powerful: You can do very advanced analysis with it.
- Lots of help online: If you get stuck, you can find videos, forums, and tutorials.

- You can use R online with: https://posit.cloud

An Introduction to Data

Let's learn how data is stored and represented in R programming.

- **Type or Mode**, refers to what kind of data we are dealing with (like numbers, text, etc.).
- Consider a list of numbers that represent the ages of family members: 43, 42, 12, 8, 5.
- All these numbers are integers; thus, we say they are the same type or mode: in this case, **integer type**.

What is “8" in R?

- “8" is inside quotes.
- **Anything inside quotes is treated as text.**
- So, in R, mode("5") is **character**, even though it looks like a number,

What is 8 in R?

- 8 has no quotes, so it is treated as a **number**

Vector in R

- A simple list of values of the same kind.

- How to Create a vector in R?

  Use the c() function, like this: **c(43, 42, 12, 8, 5)**

  - The **c() function** stands for combine.

  - If you want to give your vector a name, you do this:

    myFamilyAges <- c(43, 42, 12, 8, 5)

  - **myFamilyAges** is the name of the vector, and **<-** tells R to save the list of numbers under the name myFamilyAges.

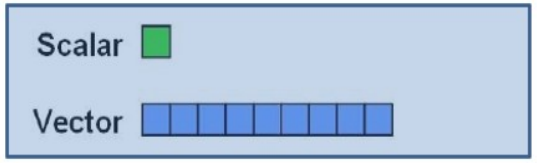
How data is organized and stored in R

These are the simplest kinds of values you can store in R.

- **Logical:**
  - Only two possible values: TRUE or FALSE
  - Internally, R often treats TRUE as 1 and FALSE as 0
- **Integer:**
  - Whole numbers like 1, 42, or -3
  - No decimal point allowed
- **Double:**
  - Numbers with decimal points, like 3.14, 0.5, or 100.0
- **Character:**
  - Text, such as "Hello" or "R is fun!“
  - Can contain letters, numbers, symbols, but treated as text, not numbers

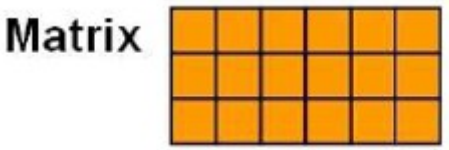
How data is organized and stored in R

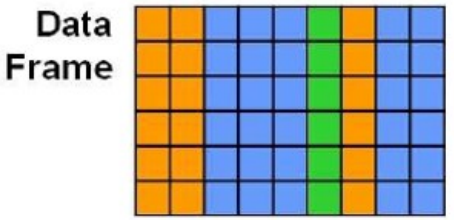
- **Scalar (Single Value):** Just one value (e.g., 5).



- **Vector:** Shown as a row of blue blocks.

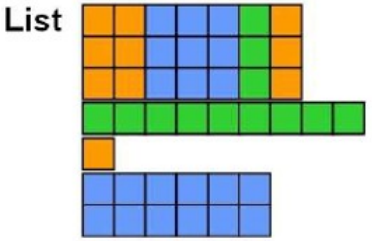
- **Matrix:**
  - A 2D table of data (rows and columns)
  - All values must be the same type.





- **Data Frame**:
  - A table, like a spreadsheet.
  - Each column can be a different type. One column can be numbers, another text, another logical.

- **List:**
  - A container that can hold anything, Numbers, Text, Vectors, Entire data frames.



Coding with R

Command	Explanation
Creating a Vector	C() e.g., c(43, 42, 12, 8, 5)
Naming/Storing a Vector	e.g. : myFamilyAges <- c(43, 42, 12, 8, 5)
Finding the sum of the Vector	sum()
Finding the Mean of the Vector	mean()
Finding the range	range()
Length of the vector	length( )
Min and Max of a vector	min() and max()

Subsetting

- **Goal:** Remove anyone younger than 30 years.
- **Data Given:** Ages in our vector: 18, 25, 31, 40, 22

```
ageNew <- age[age > 30]
```

Explanation:

- **What does age > 30 do?**
- Output: FALSE FALSE TRUE TRUE FALSE
- This is a logical vector, not the data you want.
- **What does age[age > 30] do?**
- age[age > 30], Output:31 40

Weekly Quiz

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The End